

JEE MAIN REVISION MASTER: ELECTROCHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 35 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Nernst Equation & Electrochemical Cells	25 PYQs	Electrochemical Thermodynamics
High	Conductance, Kohlrausch's Law & Faraday's Laws	10 PYQs	Conductance & Electrolysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nernst Equation & Electrochemical Cells	$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log Q, \quad \Delta G^{\circ} = -nFE^{\circ}_{\text{cell}}, \quad E^{\circ}_{\text{cell}} = \frac{0.0591}{n} \log K$
Conductance, Kohlrausch's Law & Faraday's Laws	$\Lambda_m = \frac{1000 \kappa}{M}, \quad \alpha = \frac{\Lambda_m}{\Lambda_m^{\circ}}, \quad w = \frac{M}{nF}, \quad F \approx 96500 \text{ C/mol}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nernst Equation & Electrochemical Cells (Very High Yield — 25 PYQs)

Core Principle: Nernst equation at 298 K: $E_{\text{cell}} = E^{\circ}_{\text{cell}} - (0.0591/n) \log Q$. Standard cell potential $E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$. Thermodynamic link: $\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}} = -2.303 RT \log K$. More positive reduction potential indicates stronger oxidizing power.

■ **Caution / JEE Pitfall:** Electrode potentials are intensive properties and CANNOT be added directly when combining half-cells; you must combine Gibbs energies: $\Delta G_3 = \Delta G_1 + \Delta G_2 \implies n_3 E_3^{\circ} = n_1 E_1^{\circ} + n_2 E_2^{\circ}$.

■ **Speed Trick:** At equilibrium, cell potential drops to zero ($E_{\text{cell}} = 0$) and the battery is completely discharged!

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Given below are the half-cell reactions: $\text{Mn}^{2+} + 2e^- \rightarrow \text{Mn}$ ($E^{\circ} = -1.18\text{V}$) $2\text{Mn}^{3+} + e^- \rightarrow \text{Mn}^{2+}$ ($E^{\circ} = 1.51\text{V}$)" [Ans: N/A]
- [2026 Shift] Q#4 (MCQ): "During the electrolysis of acidified water, 16 g of O_2 gas is formed at anode. The volume of H_2 gas liberated at c..." [Ans: A]
- [2025 Shift] Q#1 (MCQ): "The correct statement/s about Galvanic cell is/are (a) Current flows from cathode to anode (b) Anode is positive terminal (c) If $E_{\text{cell}} = 0$..." [Ans: N/A]

3.2 Conductance, Kohlrausch's Law & Faraday's Laws (High Yield — 10 PYQs)

Core Principle: Conductivity $\kappa = G \cdot G^* = (1/R) \left((l/A) \right)$. Molar conductivity $\Lambda_m = (1000\kappa / M)$. Kohlrausch's law of independent migration: $\Lambda_m^\circ = v_+ \lambda_+^\circ + v_- \lambda_-^\circ$. Degree of dissociation $\alpha = (\Lambda_m / \Lambda_m^\circ)$. Faraday's law of electrolysis: $w = (M \cdot I \cdot t / n F)$.

■ **Caution / JEE Pitfall:** As dilution increases, specific conductivity κ DECREASES (fewer ions per unit volume), but molar conductivity Λ_m INCREASES!

■ **Speed Trick:** 1 Faraday of charge (96500 C) deposits exactly 1 gram-equivalent of any substance at an electrode.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The conductivity of centimolar solution of KCl at 298 K is $0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing ..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): " $\Lambda_m(\text{m}(\text{NH}_4)\text{OH})^0$ is equal to _____." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Match the following select the correct option for the quantity of electricity, in Cmol^{-1} required to deposit various metals at cathode..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The conductivity of centimolar solution of KCl at 298 K is $0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing the solution at 298 K is 60Ω . The value of cell constant (G^*) is _____.

- (A) 3.28 cm^{-1}
- (B) 1.26 cm^{-1}
- (C) 3.34 cm^{-1} [Correct]
- (D) 1.34 cm^{-1}

Correct Answer: C

Solution Analysis: Conductivity is related to conductance and cell constant by $\kappa = G \times G^*$ where $G = (1/R)$. So, $\kappa = (1/R) \times G^*$. This gives $G^* = \kappa \times R$. Given, $\kappa = 0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and $R = 60 \Omega$. So, $G^* = 0.021 \times 60 = 1.26 \text{ cm}^{-1}$. So, the correct option is 1.26 cm^{-1} Option B.

Benchmark Problem 2: KCET 2026 (Q#2)

Given below are the half-cell reactions: $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ ($E^\circ = -1.18 \text{ V}$) $2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+})$ ($E^\circ = +1.51 \text{ V}$) The E_{cell}^0 for $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ will be _____.

- (A) -2.69 V , the reaction will not occur (Non-Spontaneous)
- (B) -2.69 V , the reaction will occur (Spontaneous)
- (C) -0.33 V , the reaction will not occur (Non-Spontaneous)
- (D) -0.33 V , the reaction will occur (Spontaneous)

Correct Answer: None

Solution Analysis: For the reaction $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ one part of Mn^{2+} is getting reduced to Mn: $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ $E^\circ = -1.18 \text{ V}$. So this acts as the cathode reaction. The other part of Mn^{2+} is getting oxidised to Mn^{3+} : $2\text{Mn}^{2+} \rightarrow 2\text{Mn}^{3+} + 2\text{e}^-$. This is the reverse of $2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+})$ $E^\circ = +1.51 \text{ V}$. So the oxidation potential will be $E_{\text{ox}}^0 = -1.51 \text{ V}$. Now, $E_{\text{cell}}^0 = E_{\text{cathode}}^0 + E_{\text{anode}}^0$